QRNG comparison report

noise_ibm_fez_20260707-135456_processed.txt vs bell_ibm_fez_20260707-134857_processed.txt
Generated 2026-07-08 12:45

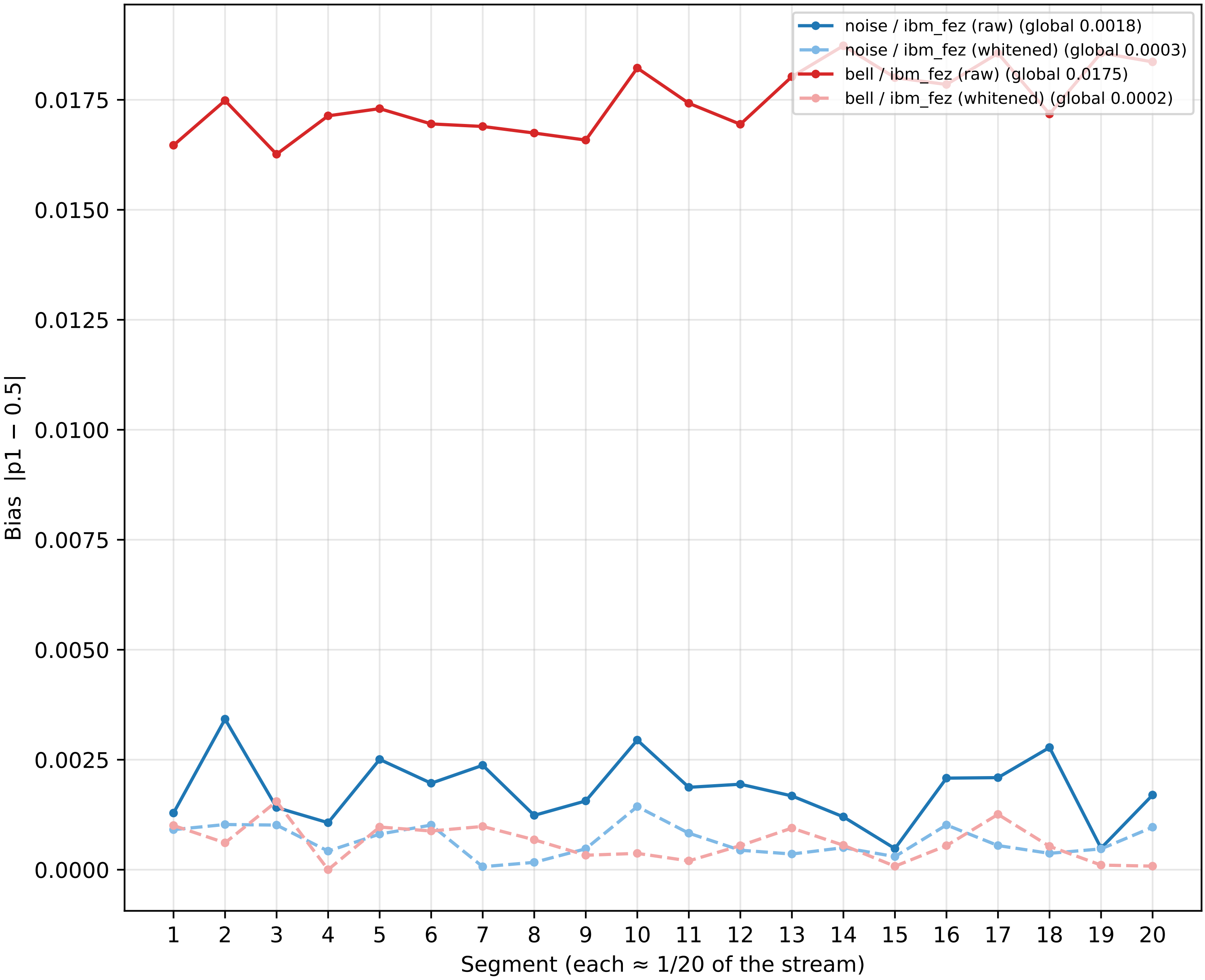
Bits used per file: ALL | NIST bits/partition: 1,000,000 | ML windows [8, 16, 24], 8 folds | Markov orders [1, 2, 4]

Metric	noise/fez raw	noise/fez white	bell/fez raw	bell/fez white
Total bits	10,540,000	7,905,000	10,016,243	7,512,184
Global bias |p1-0.5|	0.0018	0.0003	0.0175	0.0002
Min-entropy / bit	0.9949	0.9992	0.9504	0.9994
NIST pass rate	0.62	0.99	0.45	1.00
NIST passed / run	37/60	67/68	27/60	68/68
Next-bit verdict	PASS	PASS	PASS	PASS
Next-bit worst AUC	0.5013	0.5000	0.5003	0.5018
Markov verdict	PASS	PASS	PASS	PASS
Markov max excess	0.0012	0.0013	0.0006	0.0012

All four streams side by side: each input file contributes its RAW stream and its WHITENED (SHA-512, 75% kept) stream. Lower bias is better; min-entropy near 1.0 is better;
NIST pass rate near 1.0 is better; next-bit / Markov verdict PASS means no detectable structure (accuracy ~0.5 with p >= 0.01 and a 95% CI that includes 0.5).

Bias per segment (20 splits)

Local balance along each stream — all four streams overlaid



Each stream is cut into 20 equal segments; the plotted value is how far that segment's fraction of ones sits from 0.5. Solid lines = raw, dashed = whitened. A good source is LOW and FLAT:
no segment drifts and there is no rising/falling trend. Whitened lines should sit near the bottom regardless of how their raw counterpart behaves — that is the extractor removing residual imbalance. Reading tip: compare the two SOLID (raw) lines to judge the hardware; the dashed lines show what post-processing recovers.

Bias per segment — data (|p1 − 0.5| per split)

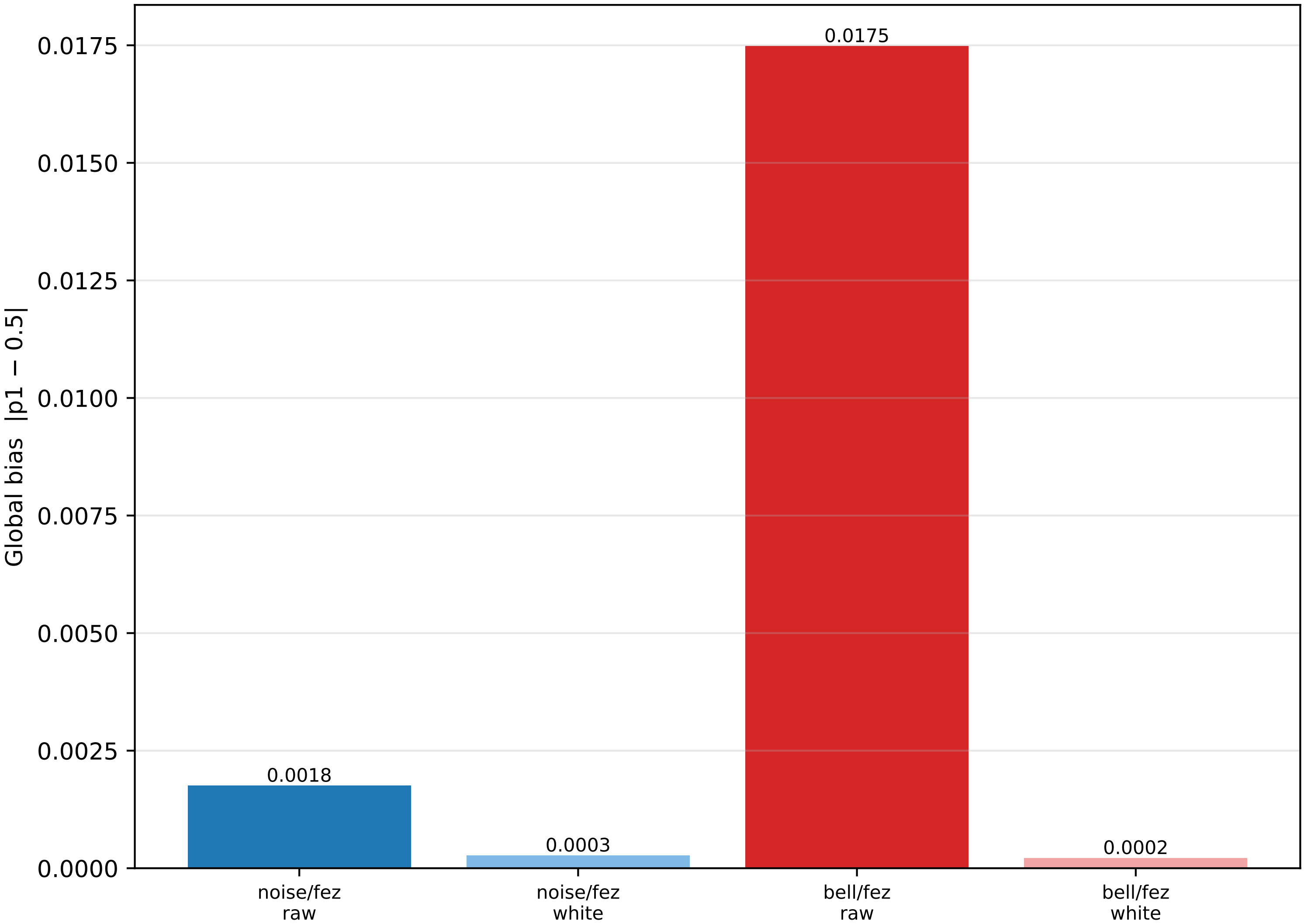
Data table (same numbers as the chart)

Split	noise/fez raw	noise/fez white	bell/fez raw	bell/fez white
1	0.0013	0.0009	0.0165	0.0010
2	0.0034	0.0010	0.0175	0.0006
3	0.0014	0.0010	0.0163	0.0016
4	0.0011	0.0004	0.0171	0.0000
5	0.0025	0.0008	0.0173	0.0010
6	0.0020	0.0010	0.0170	0.0009
7	0.0024	0.0001	0.0169	0.0010
8	0.0012	0.0002	0.0167	0.0007
9	0.0016	0.0005	0.0166	0.0003
10	0.0029	0.0014	0.0182	0.0004
11	0.0019	0.0008	0.0174	0.0002
12	0.0019	0.0004	0.0169	0.0005
13	0.0017	0.0004	0.0180	0.0009
14	0.0012	0.0005	0.0187	0.0006
15	0.0005	0.0003	0.0180	0.0001
16	0.0021	0.0010	0.0178	0.0005
17	0.0021	0.0005	0.0186	0.0013
18	0.0028	0.0004	0.0172	0.0005
19	0.0005	0.0005	0.0186	0.0001
20	0.0017	0.0010	0.0184	0.0001

Same values as the chart above. Lower is better; look for any column that is systematically larger (a more biased stream) or any single spiking row (localised imbalance).

Global bias & min-entropy — raw vs whitened

Whole-stream imbalance for all four streams



Whole-stream bias (one number per stream). The whitened bars should be far lower than their raw counterparts — direct evidence the SHA-512 extractor removes imbalance while keeping
75% of the data. Min-entropy per bit (table) converts bias to 'how many bits of true randomness per output bit'; 1.0 is ideal.

Global bias & min-entropy — data

Data table (same numbers as the chart)

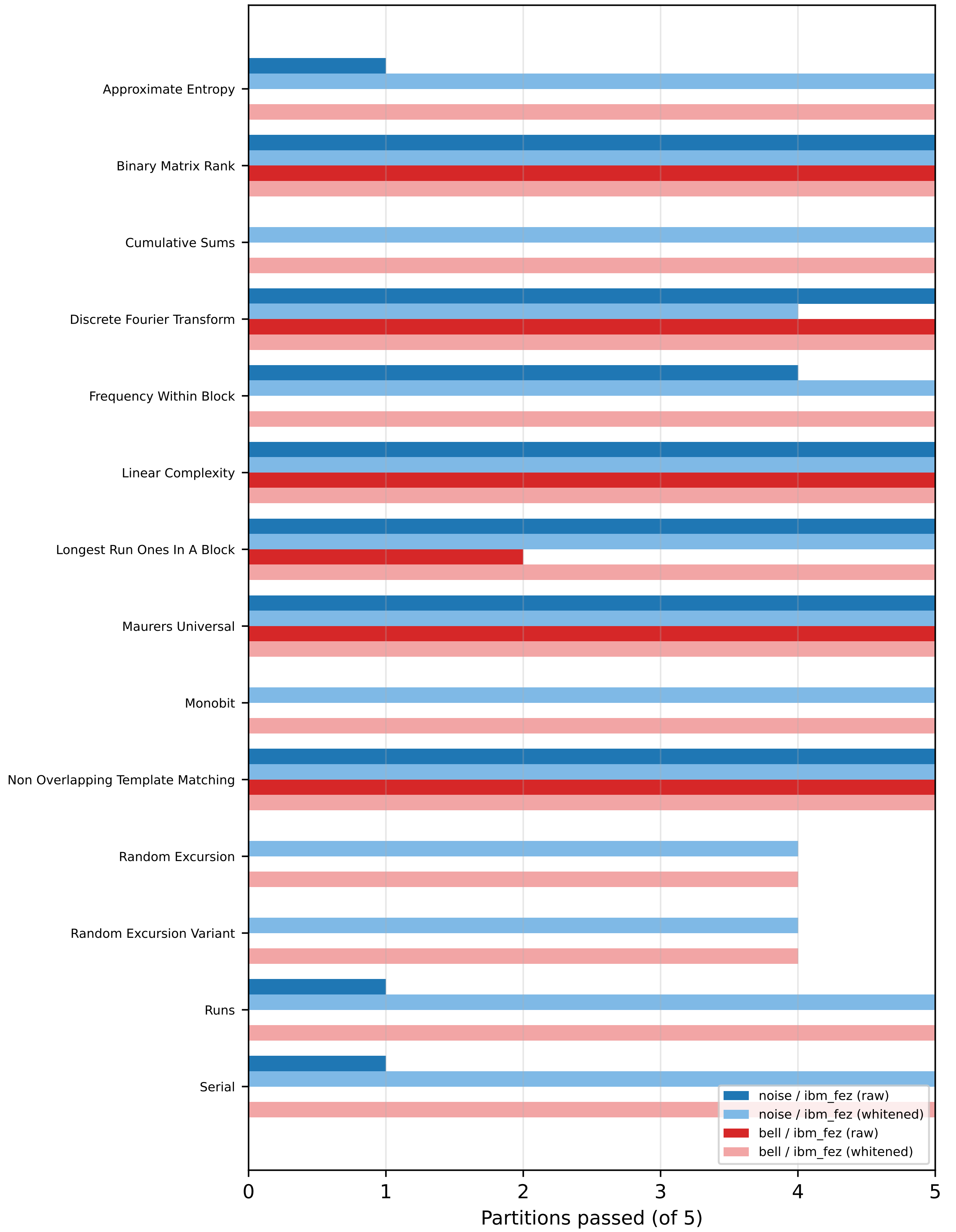
Metric	noise/fez raw	noise/fez white	bell/fez raw	bell/fez white
Global bias	0.0018	0.0003	0.0175	0.0002
Min-entropy/bit	0.9949	0.9992	0.9504	0.9994
Total bits	10,540,000	7,905,000	10,016,243	7,512,184

Same numbers as the chart. Compare each raw stream with its whitened version to quantify how much structure the extractor removed.

NIST SP 800-22 — partitions passed per test

Out of 5 partitions; all four streams grouped per test

Overall pass rate — noise/fez raw: 0.62   noise/fez white: 0.99   bell/fez raw: 0.45   bell/fez white: 1.00



The full NIST SP 800-22 battery runs on 5 partitions of each stream; a bar of 5 means the test passed in every partition (p > 0.01 each). Longer is better. Tests that need more bits
than a partition provides are skipped by the library (so the test list can differ between streams if bit counts differ). Whitened streams should reach 5 across the board.

NIST SP 800-22 — data (partitions passed / run)

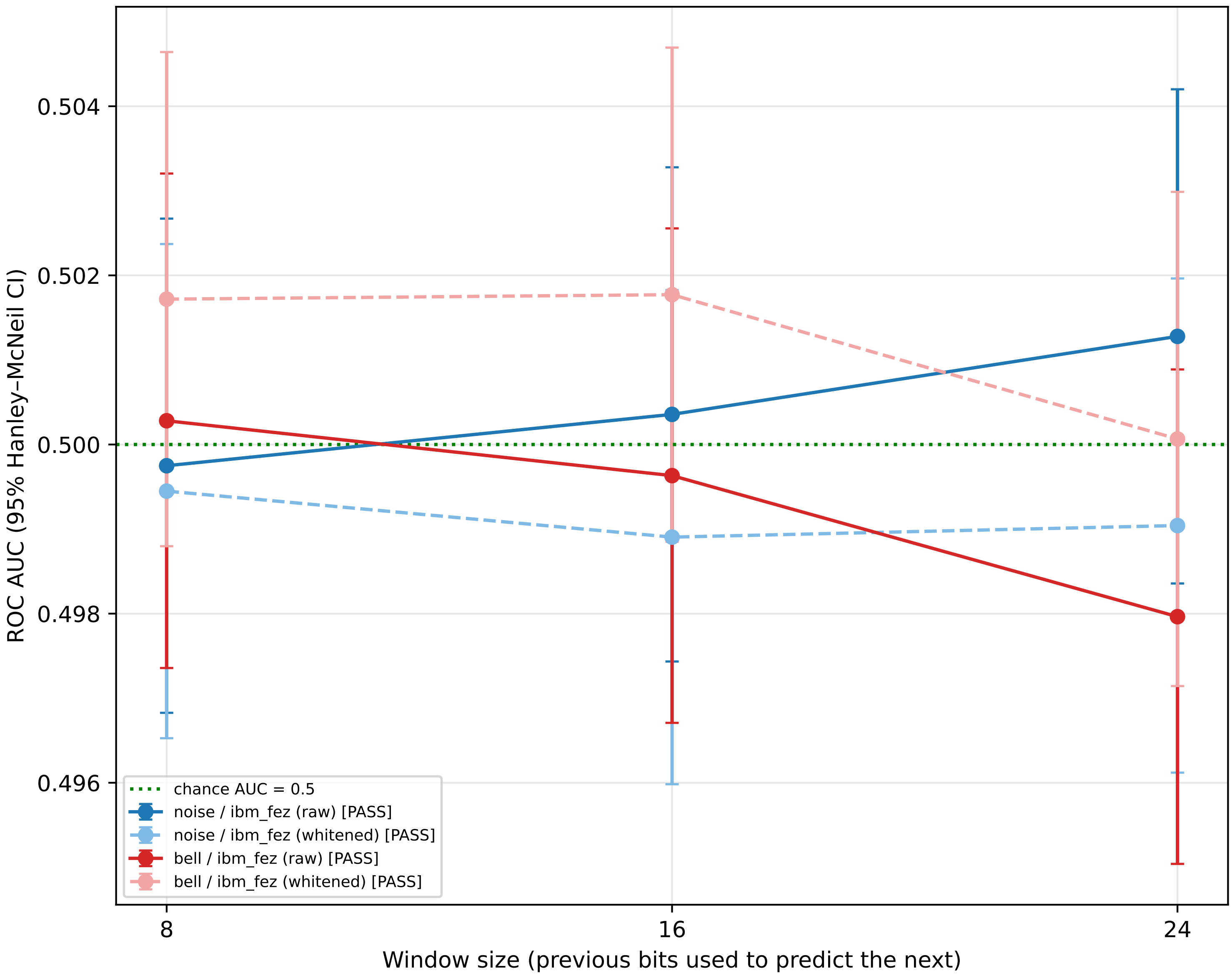
Data table (same numbers as the chart)

NIST test	noise/fez raw	noise/fez white	bell/fez raw	bell/fez white
Approximate Entropy	1/5	5/5	0/5	5/5
Binary Matrix Rank	5/5	5/5	5/5	5/5
Cumulative Sums	0/5	5/5	0/5	5/5
Discrete Fourier Transform	5/5	4/5	5/5	5/5
Frequency Within Block	4/5	5/5	0/5	5/5
Linear Complexity	5/5	5/5	5/5	5/5
Longest Run Ones In A Block	5/5	5/5	2/5	5/5
Maurers Universal	5/5	5/5	5/5	5/5
Monobit	0/5	5/5	0/5	5/5
Non Overlapping Template Match	5/5	5/5	5/5	5/5
Random Excursion	—	4/4	—	4/4
Random Excursion Variant	—	4/4	—	4/4
Runs	1/5	5/5	0/5	5/5
Serial	1/5	5/5	0/5	5/5
OVERALL pass rate	0.62	0.99	0.45	1.00

Same values as the chart. '4/5' means the test passed in 4 of the 5 partitions. '—' means the test was not eligible for that stream's length.

Next-bit predictability — strict, bias-robust (AUC)

AUC vs window size; 0.5 = no learnable structure



A logistic-regression model predicts each bit from the previous W bits, 8-fold time-ordered CV, for W in [8, 16, 24]. We use ROC AUC as the primary signal because it is BIAS-ROBUST: a biased-but-independent stream still gives AUC ~ 0.5, whereas plain accuracy would rise just from guessing the majority bit. Structure is flagged only if a window's AUC 95% CI lies ABOVE 0.5 (or accuracy beats the majority baseline). Verdict in the legend is the strict AND over all windows. (Raw bias itself is reported separately in the bias analysis.)

Next-bit predictability — data (per stream × window)

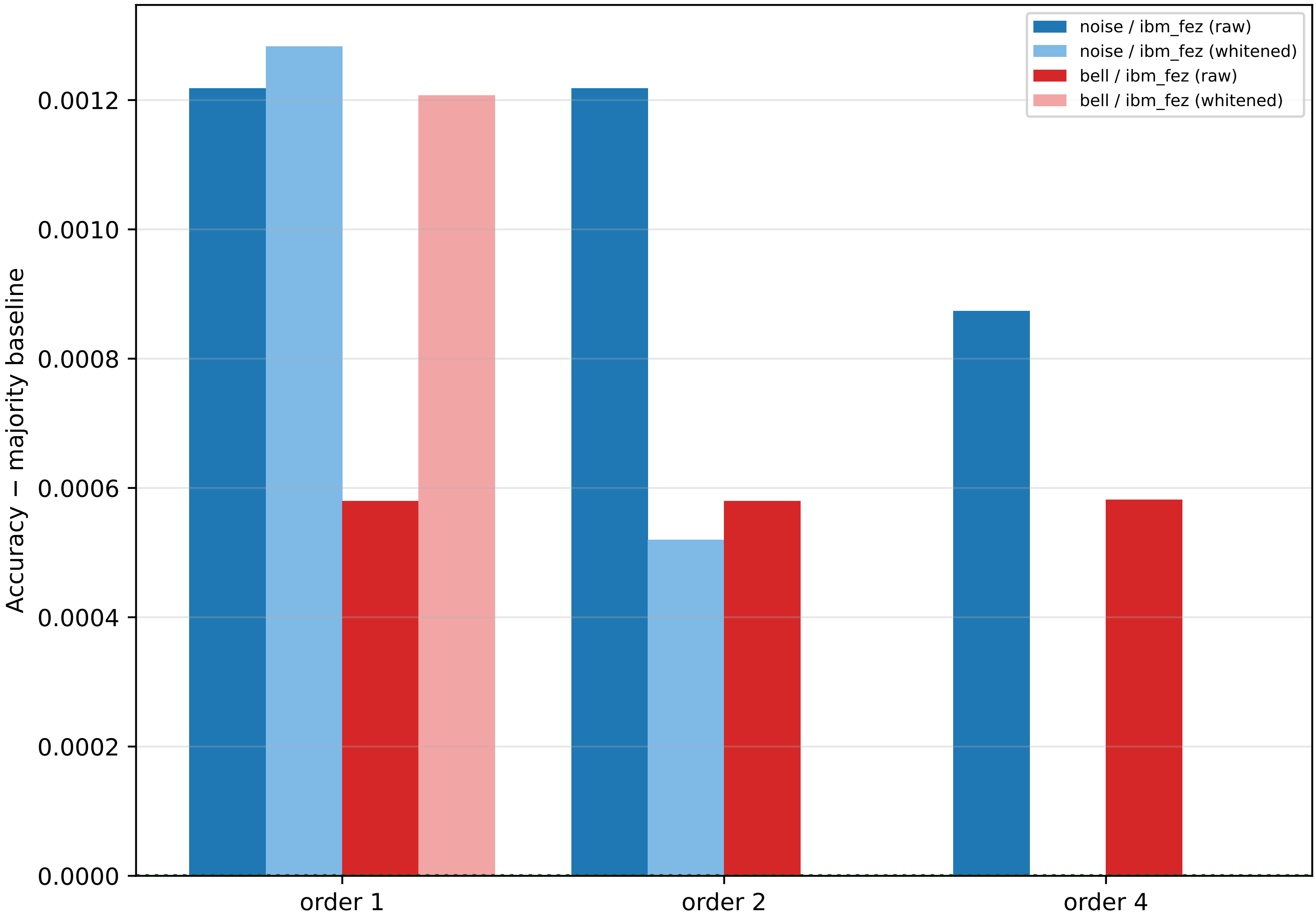
Data table (same numbers as the chart)

Stream	Window	Accuracy	Majority base	AUC	AUC 95% CI	p	pass
noise/fez raw	8	0.5003	0.5003	0.4997	[0.497,0.503]	0.415	PASS
noise/fez raw	16	0.4992	0.5003	0.5004	[0.497,0.503]	0.726	PASS
noise/fez raw	24	0.5017	0.5003	0.5013	[0.498,0.504]	0.097	PASS
noise/fez white	8	0.4990	0.5009	0.4994	[0.497,0.502]	0.772	PASS
noise/fez white	16	0.4990	0.5009	0.4989	[0.496,0.502]	0.781	PASS
noise/fez white	24	0.4999	0.5009	0.4990	[0.496,0.502]	0.518	PASS
bell/fez raw	8	0.5160	0.5174	0.5003	[0.497,0.503]	0.000	PASS
bell/fez raw	16	0.5160	0.5174	0.4996	[0.497,0.503]	0.000	PASS
bell/fez raw	24	0.5159	0.5174	0.4980	[0.495,0.501]	0.000	PASS
bell/fez white	8	0.5011	0.5007	0.5017	[0.499,0.505]	0.194	PASS
bell/fez white	16	0.5023	0.5007	0.5018	[0.499,0.505]	0.041	PASS
bell/fez white	24	0.4998	0.5007	0.5001	[0.497,0.503]	0.575	PASS

'Majority base' = accuracy obtainable by always guessing the more common bit (pure bias). A window PASSES when AUC's CI includes/sits at 0.5 AND accuracy does not significantly exceed the majority baseline.

Markov dependency — multi-order (bias-robust)

Accuracy ABOVE the majority baseline; 0 = no dependency



A k-th order Markov predictor guesses the next bit from the previous k bits. We plot how far its accuracy exceeds the MAJORITY (bias-only) baseline, so pure bias reads as ~0 — only genuine short-range DEPENDENCY lifts a bar above the green line. Whitened streams should sit at 0 at every order. (Overall bias is reported separately.)

Markov dependency — data (accuracy, baseline, excess per order)

Data table (same numbers as the chart)

Stream	o1 acc	o2 acc	o4 acc	base	o1 excess	o2 excess	o4 excess	verdict
noise/fez raw	0.5037	0.5037	0.5034	0.5025	0.0012	0.0012	0.0009	PASS
noise/fez white	0.5016	0.5008	0.4984	0.5003	0.0013	0.0005	-0.0019	PASS
bell/fez raw	0.5176	0.5176	0.5176	0.5170	0.0006	0.0006	0.0006	PASS
bell/fez white	0.5018	0.4997	0.4993	0.5006	0.0012	-0.0009	-0.0013	PASS

'base' = majority baseline (pure bias). 'excess' = accuracy − baseline; a positive, statistically significant excess at any order flags dependency and fails the Markov verdict.

Conclusions

Strict per-stream verdict and raw-hardware comparison

Stream	Bias	NIST	Next-bit	Markov	OVERALL
noise/fez raw	0.0018	low	PASS	PASS	FAIL
noise/fez white	0.0003	OK	PASS	PASS	PASS
bell/fez raw	0.0175	low	PASS	PASS	FAIL
bell/fez white	0.0002	OK	PASS	PASS	PASS

Hardware (raw) comparison: lower global bias = noise / ibm_fez (raw) (0.0018 vs 0.0175).

Strict next-bit = multi-window logistic-regression CV with Wilson CIs; a single predictable window fails the stream. Markov spans orders [1, 2, 4].

Note: whitened streams typically PASS everything — a good extractor whitens any high-entropy source — so judge the HARDWARE on the RAW rows, and read the whitened rows as 'what post-processing delivers'.

OVERALL PASS requires: NIST pass rate >= 0.90, next-bit verdict PASS, and Markov verdict PASS. This is a guide, not a formal certification; sample size and calibration still matter.